

Project Proposal

Travel Itinerary Planning System

Component 2: Trip Wizard, AI Itinerary Generator & Export

Field	Details
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1. Executive Summary

This proposal describes the Travel Itinerary Planning System — the second major component of Tour Ceylon. The system helps international and local visitors plan Sri Lankan holidays through a guided multi-step wizard that turns mood preferences, attraction selections, budget constraints, and hotel choices into a personalised day-by-day timed itinerary with PDF export.

Unlike generic travel blogs or static templates, Tour Ceylon uses curated Sri Lankan tourism datasets (937 hotels, 350 attractions, 417 travel agencies, 2,906 licensed guides, and 126 local businesses) combined with a transparent rule-based itinerary engine. The Travel component shares authentication and backend infrastructure with the SOS Emergency Response System, enabling a single login across planning and safety features.

2. Background & Motivation

- Planning a Sri Lanka trip involves scattered information across websites, guidebooks, and social media — attractions, hotels, budgets, local services, and licensed guides are rarely in one place.
- Tourists struggle to align mood preferences, number of days, multiple destinations, and budget into a coherent day-by-day schedule.
- Existing tools offer generic templates rather than personalised plans built from real Sri Lankan catalog data and user selections.
- There is no integrated path from trip planning to emergency assistance — Tour Ceylon addresses both through a unified platform.

3. Problem Statement

Tourists visiting Sri Lanka lack a single digital tool that combines mood-based attraction discovery, budget-aware hotel matching, rule-based day-by-day itinerary generation, local business and guide recommendations, and exportable trip documents. Manual planning is time-consuming, error-prone (e.g. scheduling more destinations than available days), and does not leverage structured tourism datasets already maintained by authorities and industry bodies.

4. Project Objectives

#	Objective	Success Indicator
O1	Provide mood-based attraction browsing and selection	350 attractions filterable by 10 mood tags
O2	Guide users through budget splitting and hotel selection	Budget pie chart + per-destination hotel picker within budget
O3	Generate personalised day-by-day timed itineraries	Rule-based engine produces JSON with activities and clock times
O4	Recommend local businesses, agencies, and licensed guides	SQL-filtered lists by trip destinations; save to trip
O5	Export complete trip plan as downloadable PDF	ReportLab PDF with schedule, budget, stays, saved references
O6	Persist trip history for returning users	JWT-authenticated trips stored in PostgreSQL
O7	Bridge to SOS component from same account	Sidebar link opens Tourist SOS PWA with JWT handoff

5. Proposed Solution

The Travel system follows a six-step wizard architecture backed by a rule-based itinerary engine and curated PostgreSQL datasets.

5.1 Six-step trip wizard

Step	Page	User action	Data saved
1	Attractions	Pick moods, select places, set days	user_trip_input: moods, attraction IDs, days
2	Budget	Enter total budget; view/customise split	budget_split: food/stay/shopping/transport
3	Accommodation	Pick room type + hotel per city	user_trip_input: hotels, room_type
4	Itinerary	Set start time; click Generate	generated_itinerary: timed JSON plan
5	Discover	Browse/save shops, agencies, guides	saved_references per trip
6	Export	Download PDF	PDF from itinerary + budget + references

5.2 Itinerary generation pipeline

- **trip_plan_service.py** — splits days across cities in consecutive blocks; assigns attractions and hotels to each day (no city bouncing).
- **ai_service.py** — adds clock times using mood lookup tables, category/name hints, and fixed activity gaps (2–3 hours; skip lunch slot). Despite the filename, this is rule-based — not OpenAI or machine learning.
- **generated_itinerary table** — persists JSON plan; replaced on each regeneration.
- **pdf_service.py** — renders schedule, trip budget, selected stays, and saved references.

5.3 Room type pricing model

Hotels store nightly price ranges (min A, max B). Room type selection applies a transparent formula: Single = A, Double = (A+B)/2, Family = B. Display uses approximate formatting (e.g. ~15,000 LKR / night).

6. Methodology

The itinerary engine is fully deterministic — no model training, no Random Forest, no neural networks. Quality is evaluated through manual review and a 12-step automated API smoke test.

Phase	Activity	Deliverable
1. Data seeding	Import Excel/CSV tourism datasets into PostgreSQL	937 hotels, 350 attractions, guides, agencies, businesses
2. Auth & trip API	Register/login, trip-input, budget split endpoints	JWT-secured trip CRUD
3. Accommodation	Hotel filtering by destination + budget + room type	Accommodation page with Single/Double/Family selector
4. Itinerary engine	Day allocation, time scheduling, JSON output	trip_plan_service + ai_service

Phase	Activity	Deliverable
5. Discover & export	Business/agency/guide APIs, saved references, PDF	Discover page + Export page
6. Frontend wizard	React 6-step UI with Recharts budget chart	frontend/travel on port 5180
7. Testing	verify_seed.py + test_api_flow.py (12 steps incl. PDF)	Seed counts + smoke test pass

7. Technology Stack

Layer	Technology	Purpose
Backend	Flask 3.0.3	Unified REST API (shared with SOS)
Frontend	React 18 + Vite 5	Six-step trip wizard
Database	PostgreSQL (travel_app)	Catalogs + user trips + itineraries
ORM	SQLAlchemy 2.0	Data models
Auth	Flask-JWT-Extended + optional Google OAuth	User accounts
Data import	Pandas + openpyxl	Excel/CSV seeding
PDF	ReportLab	Itinerary PDF export
Charts	Recharts	Budget pie chart
Itinerary logic	Python rule engine	Day/time scheduling (not ML)

8. Data Requirements

Dataset	File	Rows	Provenance
Hotels	accomadation.xlsx	937	Real hotel catalog
Attractions	attractions.xlsx	350	Real catalog + images
Businesses	business_directory.xlsx	126	Local shops & services
Travel agencies	travel_agencies.xlsx	417	Licensed agencies
Tourist guides	tourist_guides.csv	2,906	srilanka.travel registry
User trips	Runtime input	Dynamic	Created per authenticated session
Heuristic tuning	MOOD_START_HOURS in code	N/A	Assumed start-time mappings

9. Scope

In scope

- User registration, login, optional Google Sign-In
- Mood-based attraction browse and multi-select (1–3 moods)
- Trip constraint: unique destinations cannot exceed trip days
- Budget splitting (default 25/35/15/25 or custom percentages)
- Per-destination hotel selection with room type pricing
- Rule-based timed itinerary generation and regeneration
- Discover tab: businesses, agencies, licensed guides + save to trip
- PDF export with budget, stays, schedule, saved references
- Trip history and SOS bridge link

Out of scope (future work)

- GPS/route optimisation between attractions (no lat/lng on attractions)
- Opening hours and traffic-aware scheduling
- ML-based recommendation ranking (current: SQL filters + heuristics)
- Real-time hotel availability and booking
- Multi-currency live exchange rates (fixed LKR/USD rate used)
- Per-day start time override in UI
- Docker Compose deployment packaging

10. Key Features

Feature	Description
Mood-based discovery	10 moods; filter 350 attractions by user preference
Smart day allocation	Consecutive city blocks — no back-and-forth routing
Budget visualisation	Interactive pie chart with customisable category split
Room type pricing	Single/Double/Family mapped to hotel price range formula
Timed activities	Mood/category-aware start times with lunch gap
Local recommendations	Businesses, agencies, guides filtered by destinations

Feature	Description
PDF export	Professional trip document with all saved selections
Unified login	Same account opens SOS PWA via JWT bridge

11. Expected Outcomes & Impact

- Tourists receive personalised Sri Lankan itineraries built from real catalog data.
- Planning time reduced from hours of research to a guided 10-minute wizard.
- Budget transparency through visual splitting and hotel filtering within stay budget.
- Local economy support via discovery of businesses, agencies, and licensed guides.
- Exportable PDF serves as a shareable trip document for travel companions or agents.
- Foundation for future enhancements (GPS routing, ML recommendations, live booking).

12. Proposed Timeline

Week	Milestone
1–2	Database schema, data seeding, attraction/hotel import
3–4	Auth, trip-input API, budget split, accommodation filtering
5–6	Itinerary engine: day allocation + time scheduling
7–8	React wizard UI: all six steps + Recharts budget chart
9	Discover tab, saved references, PDF export
10	Room type pricing, SOS bridge, testing, documentation

13. Conclusion

The Travel Itinerary Planning System transforms scattered Sri Lankan tourism information into a coherent, personalised trip plan through a transparent rule-based engine and curated datasets. Combined with the SOS Emergency Response System under the Tour Ceylon umbrella, it offers tourists an end-to-end digital companion — from dreaming and planning to staying safe on the ground.

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